

MAT1002: Calculus II L02

Quiz 4 — 2025-2026, Term 2

Wednesday, 22 Apr 2026

1. (a) (3 points) Evaluate the iterated integral $\int_0^1 \int_x^1 \sin(y^2) dy dx$.
- (b) (7 points) Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy -plane, and inside the cylinder $x^2 + y^2 = 2x$.

Solution:

- (a) Since $\sin(y^2)$ is continuous, we have

$$\int_0^1 \int_x^1 \sin(y^2) dy dx = \int_0^1 \int_0^y \sin(y^2) dx dy = \int_0^1 y \sin(y^2) dy = \frac{1 - \cos 1}{2}.$$

- (b) Note that $x^2 + y^2 = 2x \Leftrightarrow (x - 1)^2 + y^2 = 1$, which is a circle, say D , centered at $(1, 0)$ with radius 1. The volume is

$$V = \iint_D \int_0^{x^2+y^2} dz dA.$$

Let $x = 1 + r \cos \theta$, $y = r \sin \theta$, $r \geq 0$, $\theta \in [0, 2\pi)$. So $x^2 + y^2 = r^2 + 2r \cos \theta + 1$, and

$$V = \int_0^{2\pi} \int_0^1 (r^2 + 2r \cos \theta + 1) r dr d\theta = \int_0^{2\pi} \left(\frac{1}{4} + \frac{2}{3} \cos \theta + \frac{1}{2} \right) d\theta = \frac{3\pi}{2}.$$

2. (a) (4 points) Use cylindrical coordinates to evaluate the integral

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2) dz dy dx.$$

- (b) (6 points) Evaluate $\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV$, where B is the closed ball centered at the origin with radius 1.

Solution:

- (a) In cylindrical coordinates (r, θ, z) , $x = r \cos \theta$, $y = r \sin \theta$, $z = z$ with $r \geq 0$, $\theta \in$

$[0, 2\pi)$. We have

$$\begin{aligned} \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2+y^2) dz dy dx &= \int_{\theta=0}^{2\pi} \int_{r=0}^2 \int_{z=r}^2 r^2 dz r dr d\theta \\ &= 2\pi \int_{r=0}^2 r^3 (2-r) dr \\ &= 2\pi \left(\frac{16}{2} - \frac{32}{5} \right) \\ &= \frac{16\pi}{5}. \end{aligned}$$

(b) In spherical polar coordinates (r, θ, ϕ) , $x = r \cos \phi \sin \theta$, $y = r \sin \phi \sin \theta$, $z = r \cos \theta$ with $r \geq 0$, $\phi \in [0, 2\pi)$, $\theta \in [0, \pi]$. We have

$$\iiint_B e^{(x^2+y^2+z^2)^{3/2}} dV = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^1 r^2 \sin \theta e^{r^3} dr d\theta d\phi = \frac{4\pi(e-1)}{3}.$$

3. (10 points) Evaluate the integral

$$\int_1^2 \int_{1/y}^y \sqrt{\frac{y}{x}} e^{\sqrt{xy}} dx dy.$$

Solution: Let $u = \sqrt{xy}$; $v = \sqrt{y/x}$. Then $x = u/v$; $y = uv$, with $u, v > 0$. The Jacobian is

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{2u}{v} > 0.$$

The boundary of D in (x, y) -coordinates consists of the following lines:

$$y = x; \quad y = -x; \quad y = 2.$$

In terms of (u, v) , they map to

$$v = 1; \quad v = -1; \quad uv = 2.$$

Thus, after the change of variables, we have

$$\int_{u=1}^2 \int_{v=1}^{2/u} 2ue^u dv du = 2 \int_1^2 e^u (2-u) du = 2e(e-2).$$